SECTION 15 / 18

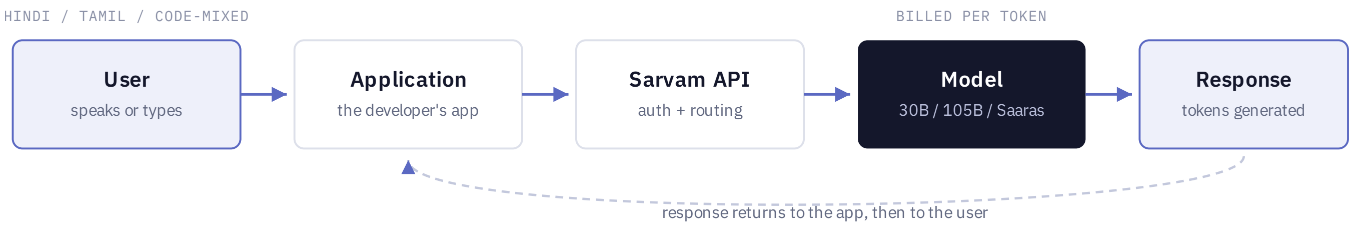
ENGINEERING LITERACY

Technical Understanding

Enough of the machinery to make good product decisions — every concept paired with the specific decision it changes, not with a textbook definition.

GOAL	CONCEPTS	DEFAULT RECOMMENDATION	CONTEXT WINDOW
Decision-grade, not ML-engineer-grade	11 + MoE	Sarvam-30B	64K on Sarvam-30B

15.1 Request lifecycle



EVERY ARROW IN THIS DIAGRAM IS A PLACE LATENCY ACCUMULATES AND COST IS INCURRED

15.2 Key concepts, and the decisions they change

CONCEPT	PLAIN-LANGUAGE EXPLANATION	WHY IT MATTERS FOR PRODUCT DECISIONS
API	The contract — endpoint plus request and response format — an app uses to talk to Sarvam's models	Determines integration effort. More APIs to choose from means more onboarding complexity, which is the whole of Section 9.
Tokens	Chunks of text, roughly three-quarters of a word, that models process and are billed on	Directly drives cost. Pricing is per million tokens, so forecasting revenue means forecasting token volume.
Latency	Time between sending a request and receiving a full response	Critical for voice agents — a slow ASR-to-TTS round trip breaks the feel of natural conversation entirely.
Inference	Running a trained model to produce an output, as opposed to training it	The ongoing cost driver after launch. Training is roughly a one-time cost; inference recurs with every single user request.
Model selection	Choosing which variant — Sarvam-30B or 105B — fits a use case	Bigger models mean better reasoning but higher cost and latency. A genuine product trade-off, not an engineering detail.
Context window	How much conversation or document history a model can see at once — 64K for Sarvam-30B	Determines whether a use case like long-document Q&A is feasible at all without extra chunking engineering.
Hallucination	The model confidently generating incorrect information	A product risk metric, not just a research one. Directly affects trust, and especially dangerous in banking and government contexts.
Evaluations	Structured tests measuring model quality on defined tasks, such as the BrowseComp and Tau2 scores cited for Sarvam-105B	How a PM decides launch readiness. Evals replace vibes as the gate.

CONCEPT	PLAIN-LANGUAGE EXPLANATION	WHY IT MATTERS FOR PRODUCT DECISIONS
Cost per request	The ₹ cost of a single API call, derived from token, character, or minute pricing	The core lever connecting technical usage to the business model in Section 13.
Uptime	Percentage of time the API is available and responding correctly	Directly shapes enterprise SLA negotiations — and the absence of a self-serve SLA tier is a gap named in Section 9.
Rate limits	Maximum requests allowed per time window, set by plan tier	The exact mechanism behind the self-serve-to-enterprise friction wall found in the teardown.

15.3 Mixture-of-Experts, and why it matters commercially

What MoE does

ARCHITECTURE

Sarvam-30B is reported to use a Mixture-of-Experts architecture in which only about **2.4B parameters are active per request**, despite the model holding 30B in total. Rather than running the whole network on every call, a router activates a small relevant subset.

Why a PM should care

PRODUCT IMPLICATION

This is *why* the model is low-latency despite being large. It can deliver near-large-model quality at lower cost and latency for most use cases — which makes it the correct default recommendation for self-serve developers, exactly as proposed in the Section 10 redesign.

THE TEST OF TECHNICAL LITERACY FOR A PM

Not being able to implement a Mixture-of-Experts router — but being able to say why the architecture choice justifies making Sarvam-30B the default recommendation, and what would have to change before recommending the 105B instead.